From temporal bilocals to bulk mesons
A concrete reconstruction interface

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Abstract

We use an existing light-front bilocal-to-meson transform to formulate the next microscopic matching problem in the Susskind programme. Internal source profiles give calculable mass mixtures, including tails invisible to a ground-state overlap alone. On the microscopic side we derive exact degree-resolved norms of the first unequal-time CAR corrections in complex SYK. Their collective zero-time curvature remains finite in double scaling. These are inputs for constructing a dressed bilocal map; the map itself remains to be determined.

1 The known bulk transformation

Itakura's boson expansion [1], especially (4.25)–(4.26) and (5.1)–(5.9), supplies canonical bilocal bosons and their meson projection. Write $P > 0$ for total light-front momentum and $0 < x < 1$ for the internal quark fraction. For real orthonormal wavefunctions,

$$b_r^\dagger(P) = \sqrt{P}\int_0^1 dx\,\varphi_r(x)B^\dagger(Px, P(1-x)), \qquad \int_0^1 \varphi_r\varphi_s\,dx = \delta_{rs}. \tag{1}$$

The Jacobian $dp_1dp_2 = P\,dPdx$ gives $[b_r(P), b_s^\dagger(P')] = \delta_{rs}\delta(P-P')$ from canonical B commutators. The bosonic pair variables are related nonlinearly to the quark bilocals, with a physical-subspace restriction; they are not simply the raw quark products at finite colour number.

We retain the mass conventions of our preceding FLZ calculation [2]: $\mathsf{M}^2 = (\tau/\pi)\mathsf{K}_0$, $\mathsf{K}_0\varphi_r = 2\pi^2\nu_r\varphi_r$, and $M_r^2 = 2\pi\tau\nu_r$, where τ is the bulk tension scale. Our external light-cone convention is $M^2 = P^+P^-$; Itakura uses $M^2 = 2P^+P^-$. Equation (1) is used as an internal projection, without importing his Hamiltonian prefactors.

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2 A source is a calculable mass mixture

For any normalized $f \in L^2(dx)$, the same construction has amplitudes $a_r = \int \varphi_r f\, dx$. In our convention its mass moments are $\sum_r |a_r|^2 (M_r^2)^j$. This is an ordinary spectral decomposition of the existing bulk operator, not yet a SYK dictionary.

Take $2x - 1 = \cos\theta$, and define internal test profiles

$$f_k(x) = \frac{\sin(k\theta)}{\sqrt{B_{kk}/2}}, \quad B_{mn} = \begin{cases} \dfrac{1}{1-(m-n)^2} - \dfrac{1}{1-(m+n)^2}, & m+n \text{ even}, \\ 0, & m+n \text{ odd}. \end{cases} \tag{2}$$

The physical Gram matrix is $B/2$. The FLZ Chebyshev representation implies $\mathsf{K}_0 \sin(k\theta) = \pi k U_{k-1}(\cos\theta)$. Consequently the full, untruncated moments are

$$\frac{\langle M^2\rangle_k}{\tau} = \frac{\pi k}{B_{kk}}, \qquad B_{kk} = 1 + \frac{1}{4k^2-1}, \tag{3}$$

$$\frac{\langle M^4\rangle_k}{\tau^2} = \frac{4k^2 C_{kk}}{B_{kk}}, \qquad C_{kk} = \int_{-1}^{1} U_{k-1}(z)^2\, dz. \tag{4}$$

For example, the first formula follows from $\int_0^\pi \sin^2(k\theta) d\theta = \pi/2$; the second is the norm squared of $\mathsf{M}^2 f_k$.

k	dominant weight	$\langle M^2\rangle/\tau$	$\langle M^4\rangle/\langle M^2\rangle^2$
1	$\lvert a_0\rvert^2 = 0.99470033$	$3\pi/4$	$32/(3\pi^2)$
2	$\lvert a_1\rvert^2 = 0.95513069$	$15\pi/8$	$512/(45\pi^2)$

The ratios are 1.08075929 and 1.15280991, respectively, both strictly larger than the single-mass value one. For $k = 1$, keeping only the first six levels misses norm 0.00020893, but misses 0.21276878 in $\langle M^4\rangle/\tau^2 = 6$. For $k = 2$ the corresponding omissions are 0.00309244 and 4.09931829 out of 40. Small norm tails need not be small relative errors in high moments.

The saved 256-mode eigenvectors give these overlaps; independent 400-node angular integration agrees within 2×10^{-12}. The source index k has not been identified with a microscopic time-derivative order. Internal x, external boost rapidity, and the relative fraction of two mesons remain different variables.

3 The exact temporal algebra to be matched

Use standard complex CAR, $\{c_i, c_j^\dagger\} = \delta_{ij}$, the infinite-temperature state $\mathrm{tr}_0 = 2^{-N}\,\mathrm{Tr}$, and $\|O\|_2^2 = \mathrm{tr}_0(O^\dagger O)$. Let

$$H = 2^p \sum_{|I|=|J|=p} J_{IJ} C_I^\dagger C_J, \quad \mathbb{E} J_{IJ} J_{KL} = v \delta_{IL} \delta_{JK}, \quad v = \frac{J_s^2 N}{p^2} \left(\frac{p!}{N^p} \right)^2. \tag{5}$$

Here J is a centred Hermitian Gaussian matrix on ordered subsets, and $C_I^\dagger$ is the ordered product of creators. Time obeys $c_i(t) = e^{iHt} c_i e^{-iHt}$, with $J_s t$ dimensionless. Define

$$D_i = [H, c_i], \qquad T_{ij} = \{D_i, c_j^\dagger\}, \qquad U_{ij} = \{D_i, c_j\}. \tag{6}$$

For finite N,

$$\{c_i(t), c_j^\dagger\} = \delta_{ij} + itT_{ij} + O(t^2), \tag{7}$$

$$\{c_i(t), c_j\} = itU_{ij} + O(t^2). \tag{8}$$

Thus unequal-time CAR cannot be replaced by equal-time scalar CAR without checking the operator corrections. For four odd operators, ordinary multiplication gives the exact identity

$$[ab, cd] = a\{b, c\}d - c\{d, a\}b + ca\{d, b\} - \{a, c\}bd. \tag{9}$$

Set $a = c_i^\dagger(t_1)$, $b = c_i(t_2)$, $c = c_j^\dagger(t_3)$, $d = c_j(t_4)$ and sum over i, j to obtain the raw temporal singlet bilocal commutator. Both mixed and anomalous anticommutators occur. At coincident times standard cancellations recover the familiar number-bilinear algebra.

To quantify the corrections, resolve the neutral Hamiltonian into orthogonal Majorana degrees $2k$: $H = \sum_k H_k$ and $w_k = \|H_k\|_2^2$. This degree is neither the $U(N)$ covariance rank of earlier notes nor the Gaussian degree in the couplings. The exact identities are

$$\sum_i \|D_i\|_2^2 = 2 \sum_k k w_k, \qquad \sum_{ij} \|T_{ij}\|_2^2 = 4 \sum_k k^2 w_k, \tag{10}$$

$$\sum_{ij} \|U_{ij}\|_2^2 = 4 \sum_k k(k-1) w_k, \qquad \sum_{ij} |\,\mathrm{tr}_0\, T_{ij}|^2 = 4 w_1.$$

These are complex-fermion versions of the operator-size counting used in infinite-temperature SYK diagnostics [3].

4 Degree counting and its double-scaling limit

For completeness, use the exterior-algebra realization of the Hilbert–Schmidt Clifford basis, with orthonormal one-forms $\sqrt{2}c_i, \sqrt{2}c_i^\dagger$. A neutral degree-$2k$ vector has k forms of each charge. Commutation with c_i on the even sector is a contraction with factor $\sqrt{2}$, giving total squared norm $2k$. The resulting odd vector has k annihilation forms and $k-1$ creation forms. Anticommutation with $c_j^\dagger$ or c_j contracts these with another factor $\sqrt{2}$. Summing all ordered contractions gives $4k^2$ or $4k(k-1)$. Orthogonality between degrees removes cross terms. Only $k=1$ reaches degree zero after two contractions, proving the last identity in (10).

The Gaussian covariance reduces each ensemble norm to a sum over single monomials. For $r = |I \cap J|$, expand each shared occupation factor as $n_i = (1+z_i)/2$, where $z_i = 2n_i - 1$. Choosing s factors z_i gives degree $2k = 2(p-r+s)$. In $2^p C_I^\dagger C_J$ every such choice has squared norm one: the factors from the disjoint ladder operators cancel $2^{2(p-r)}$. Counting pairs of subsets therefore yields

$$\frac{\mathbb{E}w_k}{v} = \binom{N}{p}\sum_r \binom{p}{r}\binom{N-p}{p-r}\binom{r}{p-k} = \binom{N}{p}\binom{p}{k}\binom{N-p+k}{k}. \tag{11}$$

The last step is Vandermonde's identity after extracting $\binom{p}{p-k}$. Impossible binomial coefficients vanish.

Put $\overline{w}_k = \mathbb{E}w_k$ and $\ell = p-k$. Then exactly

$$\frac{\overline{w}_{p-\ell}}{\overline{w}_p} = \frac{(p)_\ell^2}{\ell!(N)_\ell}, \qquad \frac{r_{\ell+1}}{r_\ell} = \frac{(p-\ell)^2}{(\ell+1)(N-\ell)} \le \frac{p^2}{(N-p)(\ell+1)}. \tag{12}$$

Here $(a)_\ell$ denotes a falling factorial and $N > p$. For $p, N \to \infty$ with $p^2/N \to \lambda > 0$, the bound supplies a summable Poisson-type majorant, including every fixed moment. The normalized deficit law tends to $e^{-\lambda}\lambda^\ell/\ell!$. Also $\overline{w}_p = v\binom{N}{p}^2 \to (J_s^2/\lambda)e^{-\lambda}$, so

$$\mathbb{E}\|H\|_2^2 \longrightarrow J_s^2/\lambda, \qquad \mathbb{E}_{\text{degree}}(k^2)/p^2 \longrightarrow 1. \tag{13}$$

The expectation labelled "degree" uses the normalized ensemble weights, not a new disorder ensemble.

5 A finite microscopic matching datum

Combining the preceding formulas gives

$$\frac{1}{N}\mathbb{E}\sum_{ij}\|T_{ij}\|_2^2 \;\longrightarrow\; 4J_s^2, \qquad \frac{1}{N}\mathbb{E}\sum_{ij}\|U_{ij}\|_2^2 \;\longrightarrow\; 4J_s^2. \tag{14}$$

The first limit is unchanged on replacing T_{ij} by $T_{ij} - \mathrm{tr}_0(T_{ij})\mathbf{1}$, because

$$\frac{\overline{w}_1}{\sum_k k^2\overline{w}_k} \le \frac{N-p+1}{p\binom{N}{p}} \longrightarrow 0. \tag{15}$$

Thus a scalar replacement discards the leading collective curvature. More precisely, for example,

$$\lim_{\mathrm{DS}}\lim_{t\to 0}\frac{1}{Nt^2}\mathbb{E}\sum_{ij}\|\{c_i(t), c_j\}\|_2^2 = 4J_s^2. \tag{16}$$

The order of limits is part of the result. It is not a uniform finite-time estimate. Nor does it determine what survives after all four terms in (9) are multiplied, summed and projected onto a particular singlet subspace.

λ, $p = 128$, $J_s = 1$	mixed norm/N	anomalous norm/N
0.25	3.98972538	3.95849537
0.50	3.97446219	3.94329182
1.00	3.92938402	3.89844955

Finite-p deviations are retained rather than reported as numerical error. The saved calculations include five exact integer-matrix Hamiltonians, 77 combinatorial cases, 54 direct monomial covariance checks, and 15 evaluations at 80-digit precision. The four-operator identity was also checked on explicit odd matrices. These validate the stated formulas; no real-time double-scaling extrapolation is used.

6 The next construction

The immediate target is the projected algebra and covariance of $S(t_1, t_2) = \sum_i c_i^\dagger(t_1)c_i(t_2)$, retaining all four terms in (9). We should identify its physical polarization and any required nonlinear dressing before applying a meson projection. The known bulk collective-action route [4] provides another concrete comparison: match its quadratic internal operator and then interaction vertices. The familiar SYK bilocal construction [5] provides useful methods, but its conformal infrared saddle does not supply this infinite-temperature complex double-scaled map. This is where the next calculation begins.

Reproducibility and scope

The bulk calculation uses the preceding stage's 256-mode wavefunctions at $a = 0$ and retains only six output eigenstates. Full source moments are evaluated analytically; omitted contributions are recorded separately. They are differences from a converged finite-basis result, not interval-certified spectral bounds.

The microscopic calculation uses normalized trace, standard CAR, centred Gaussian couplings, and the variance specified above. Exact matrix tests use Gaussian-integer entries with seed 110630; their integer bounds are checked before conversion to rational norms. The general proofs use Clifford contractions and subset combinatorics, not a fit to these small systems. No finite-gravity interaction kernel or microscopic bulk translation is inferred from these norms alone.

The canonical bulk transform and operator-size method are established literature. The formulas here apply them to the present normalization and matching question; no priority claim is made. The next test must address the full singlet projection rather than merely repeat the unprojected positive-norm calculation.

References

[1] K. Itakura, *Boson Expansion Methods in (1+1)-dimensional Light-Front QCD*, hep-th/9604032v2.

[2] V. A. Fateev, S. L. Lukyanov and A. B. Zamolodchikov, *On mass spectrum in 't Hooft's 2D model of mesons*, arXiv:0905.2280v2.

[3] D. A. Roberts, D. Stanford and A. Streicher, *Operator growth in the SYK model*, arXiv:1802.02633.

[4] M. Cavicchi, P. Di Vecchia and I. Pesando, *The Master Field of QCD_2 and the 't Hooft Equation*, hep-th/9306091.

[5] A. Jevicki, K. Suzuki and J. Yoon, *Bi-Local Holography in the SYK Model*, arXiv:1603.06246.